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Fields of Concentration:

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Secondary Field: Economic Development

Qualifying Examinations Completed:

2023: International Trade (*with distinction*), Economic Development (*with distinction*)

Dissertation Title: *Essays on International Trade, Technology, and Firms*

Committee:

Professor Peter Schott (Chair)

Professor Amit Khandelwal

Professor Samuel Kortum

Professor Teresa Fort

Education:

Ph.D., Economics, Yale University, 2027 (expected)

M.Phil., Economics, Yale University, 2024

M.A., Economics, Yale University, 2023

B.A., Mathematics and Economics (*with highest distinction*), University of Virginia, 2020

Fellowships, Honors and Awards:

Ryoichi Sasakawa Young Leaders Fellowship (SYLFF), 2023-2024

Cowles Foundation Fellowship, 2021-2026

Doctoral Fellowship, Yale University, 2021-2026

Best Undergraduate Paper Award, International Atlantic Economic Society, 2020

Teaching Experience:

Spring 2026, Teaching Assistant to Prof. Peter Schott, Decision Making with Data (MBA),
Yale University School of Management

Fall 2025, Teaching Assistant to Prof. Teresa Delgado, Long-Term Fiscal Sustainability
(undergraduate), Yale College
Summer 2024, Co-Director, Herb Scarf Summer Research Program in Economics
(undergraduate), Yale University

Research Experience:

Research Assistant to Prof. Peter Schott and Prof. Teresa Fort, Yale University, 2023-2025
Research Assistant to Prof. Paul Goldsmith-Pinkham, Yale University, Summer 2022

Work Experience:

Data Scientist, 2nd Order Solutions, 2020-2021

Publications:

“[Selective Entry in Highway Procurement Auctions](#)” (2020), *Atlantic Economic Journal*,
48, 519–533.

Working Papers:

“Offshoring and Knowledge Spillovers”, *Job Market Paper*

“Workforce Age Structure and Frontier Technology Adoption: Evidence from Newly
Released Software” with Tra Nguyen

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References:

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Dissertation Abstract

Offshoring and Knowledge Spillovers (*Job Market Paper*)

U.S. firms have relocated production to China to access lower labor costs. A by-product of this offshoring is the transfer of production knowledge to firms' affiliates or suppliers abroad. This know-how may diffuse to nearby producers, strengthening foreign producers that could later compete with U.S. manufacturers. While the rise of import competition from China has received substantial attention, less is known about the extent to which such offshoring-induced local spillovers occurred and their contribution to China's export growth.

This paper studies offshoring as a channel of know-how diffusion from the U.S. to China and quantifies its implications for U.S. welfare. I first measure offshoring in U.S.-China trade from 1997 to 2017 using U.S. Census microdata. I define offshoring as a U.S. manufacturer importing a product that it produced domestically in 1997. I measure firms' domestic production using the Longitudinal Business Database and Census of Manufactures, which I link to the universe of U.S. import transactions in the Longitudinal Foreign Trade Transactions Database. I document the share of offshored goods in U.S. manufacturers' imports from China and the value of such imports relative to U.S. manufacturing shipments. Using the buyer-supplier links in these data, I examine whether offshoring relationships differ from other trade relationships in ways consistent with know-how transmission, such as longer duration and higher annual trade value.

I next ask whether offshoring generates knowledge spillovers to nearby Chinese exporters that do not themselves receive offshoring orders. I study these spillovers across Chinese city-industry cells using a difference-in-differences design. A city-industry cell is treated when it receives its first offshoring order, with treatment timing varying across cells. My main outcome is cell-level exports to the U.S. by exporters that never receive offshoring orders. Identification relies on treated and not-yet-treated cells following similar outcome trends absent offshoring, controlling for differential trends by initial cell size and sector-city-year fixed effects. Using a local projections difference-in-differences estimator, I trace the evolution of the never-offshoring exporters' sales to the U.S. from five years before to five years after treatment of their cell. I then decompose the change in export value into contributions from Chinese exporters that continue, start, and stop exporting to the United States.

Finally, I develop a quantitative model to evaluate the welfare and policy implications of offshoring-induced knowledge spillovers. The model features U.S. firms that choose whether to offshore production to China, where offshoring lowers costs, but also transfers production knowledge that diffuses to nearby producers. Knowledge diffusion reduces U.S. firm profits by increasing competition from foreign producers, while benefiting U.S. consumers through lower prices. Because offshoring firms do not internalize the effects on other firms and consumers, their private decisions may differ from the optimum chosen by a U.S. or global social planner. I use the model to quantify this wedge, assess its welfare implications, and evaluate the effects of policies that correct it.

(Results are omitted because disclosure review has not yet been completed.)

Workforce Age Structure and Frontier Technology Adoption: Evidence from Newly Released Software, with Tra Nguyen

Workforce aging may affect economic growth not only by reducing the aggregate labor supply, but also by slowing the adoption of new technologies. This paper provides causal evidence on how local workforce age structure affects firms' adoption of newly released software. Using Lightcast job postings, we measure employer demand for 15 frontier software technologies released between 2012 and 2016, each paired with an older technology used for similar tasks. We define local age structure as the share of college-educated workers aged 50 to 69 and instrument this share with a Bartik-style measure based on historical commuting-zone age composition and national survival rates, while controlling for local labor-market characteristics. We find that within five years of a technology's release, employers in older labor markets post significantly fewer vacancies requiring the newest software skills. This adoption gap is larger for technologies harder to master. To interpret the evidence and evaluate policy counterfactuals, we develop a model in which workers decide whether to retrain for frontier technologies and firm adoption depends on the supply of frontier-technology skills. The model highlights a finite-horizon mechanism: older workers have less time to recover retraining costs, making late-career skill investment less attractive. Counterfactual exercises show that different sources of aging have different implications for adoption. Aging driven by weak inflows of young workers, such as low fertility or low migration, slows technology diffusion, while policies that extend working lives, such as delayed retirement, can partly offset these lower adoption rates by raising older workers' incentives to acquire new skills.